

On the Traffic Light Timing Setting Problem *

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Abstract

This paper studies the traffic light timing setting problem. The traffic network of a city can be modeled by a graph in which vertices represent the intersections of roads and edges represent the segments of roads. The traffic light problem is concerned with the setting of the starting time of the green light for each traffic light such that the total waiting time of the traffic in the traffic network is minimized. We consider two versions of the traffic light problem: *simplified* and *realistic*. In the simplified version, there is only one traffic light on each vertex, while in the realistic version, the number of traffic lights on each vertex is equal to its degree.

In this paper, we first prove that the traffic light problem is NP-hard for the traffic light cycle $T \geq 3$, that the traffic light problem on a planar graph with $T = 2$ can be solved by the algorithm designed for the weighted maximum cut problem, whose time complexity is $O(n^{\frac{3}{2}} \log n)$, and that the two-phase setting problem (a special case of the traffic light problem) on a planar graph can also be solved in the same time complexity. Then, we discuss some properties of the traffic light problem. Using these properties, we design an algorithm to find the minimum penalty of a polygon with n edges in $O(n)$ time. Finally, we present a heuristic algorithm to solve the traffic light problem and show some experiment results.

1 Introduction

Every driver who drives on a street of a city must have the following experience. After you wait for a traffic light to turn green at an intersection and start to drive toward the next intersection, the traffic light at the next intersection turns red before you arrive at it. If the traffic light system is badly designed, drivers may waste a lot of time waiting for the lights to turn green.

The traffic condition in a city is greatly affected by the timing setting of the traffic lights, i.e., when the red light and the green light is on. If the timings of the traffic lights are not set properly, drivers may spend a lot of time waiting in front of the traffic lights. To reduce the waiting time, the timings of the traffic lights should be carefully set. In this paper, we consider the following problem: decide the starting time of the green light for each traffic light in such a way that the total waiting time of the traffic in the traffic network is minimized.

The traffic light problem has been studied extensively under various assumptions. Some researchers [2, 5, 9, 11, 13] used sensors to get the flow of the traffic and used many kinds of methods to decide the timings of the traffic lights. The study on fuzzy systems

has also been applied to the traffic light problem [11, 13]. For the simpler conditions of roads, J. Favilla *et al.* [5] put a fuzzy logic controller (FLC) into the traffic controller. Some physical researchers take a different kind of view to this problem, they discuss this problem in isolated crossroads and construct mathematical models to solve this problem. [6, 7, 10] Some other researchers thought vehicle density and total traffic flow should be considered, therefore they try to solve the problem by cellular automata (CA) models. [4] One of such a CA model is the Biham, Middleton, and Levine model (BML model) [3, 12] which perhaps is the simplest model describes traffic flow in two dimensions. Furthermore, Genetic Algorithm (GA) approach is also been discussed in some recent research. [?, ?, ?, ?]

Because the general traffic light problem is of practical interest but difficult to solve, we will make some assumptions to simplify the problem:

1. Each traffic light has only two lights, green and red.
2. All traffic lights have the same cycle (period), T , i.e., T equals to the duration of one green light and one red light.
3. The green light duration is equal to the red light duration for all the traffic lights. If a car reaches an intersection and the traffic light in its direction is red, it must wait until the light turns green, and then starts to drive toward the next intersection.
4. Each car takes one unit of time to pass through one unit of distance.

The traffic light problem can be modeled as a graph problem. We can use a graph $G = (V, E)$ to represent the traffic network of a city, where each vertex in V corresponds to an intersection of some roads, and each edge (u, v) in E corresponds to the segment of a road connecting the two intersections u and v . Each edge (u, v) is associated with a nonnegative integral length (distance) $d(u, v)$. For the simplified version of the traffic light problem, we assume that there is only one traffic light at each intersection (vertex). As we assume that all traffic lights have the same cycle T and the green light duration is equal to the red light duration for all traffic lights, it is easy to see that we can use the starting time of a green

light, denoted as $t(v)$, to specify the timing setting of the traffic light at vertex v , i.e., the traffic light at v is green in the time intervals $[t(v) + q \cdot T, t(v) + (q + 1/2) \cdot T)$, for all integer q , and red in all other time intervals. For the realistic version, we assume that there is one traffic light for each edge (u, v) incident to a vertex v . Therefore, we use $t(u, v)$ to denote the starting time of a green light of the traffic light corresponding to edge (u, v) incident to v .

For the simplified traffic light problem, suppose a car reaches intersection u , waits for the green light to turn on (if it is red light when the car reaches the intersection), and then heads to intersection v . If $t(u)$ and $t(v)$ are not set properly, when the car passes intersection u during its green light interval, goes through road segment (u, v) , and reaches intersection v , the traffic light at v turns red, and hence, the car has to wait in front of the traffic light at v before heading to the next intersection. The traffic light problem is then to assign $t(v)$ for each $v \in V$ such that this kind of waiting time can be minimized.

In this paper, we formally define the network model for solving the traffic light problem. We design optimal algorithms for some special cases of the problem, prove the NP-hardness of the general problem, provide some properties of the problem, and propose a heuristic algorithm for the general problem. We also give some simulation results to show the effectiveness of our heuristic algorithm.

The rest of the paper is organized as follows. In Section 2, we first give the definition of the penalty of an edge and then give the formal definition of the traffic light problem, including the simplified version and the realistic version. In Section 3, we prove that the traffic light problem is NP-hard for the traffic light cycle $T \geq 3$. Then, we show how to solve the problem on a planar graph with $T = 2$ by the algorithm designed for the weighted maximum cut problem [14], which has a time complexity $O(n^{\frac{3}{2}} \log n)$. In Section 4, we first define the two-phase setting problem, and solve it by using the weighted maximum cut algorithm if the underlying traffic network is a planar graph. Then we discuss some properties of the traffic light problem. In Section 5, we first give an algorithm to find the minimum penalty of a polygon with n edges in $O(n)$ time. We then use the result to design a heuristic

algorithm for solving the traffic light problem. In Section 6, we give some experiment results. Finally, the paper concludes with Section 7.

2 The Definition of the Traffic Light Problem

A traffic network is represented by a graph $G = (V, E)$, where each edge (u, v) in E corresponds to a segment of a road and each vertex v in V corresponds to the intersection of some roads. Note that we assume that each road (segment) is bi-directional. Each edge (u, v) is also associated with a nonnegative integral distance, $d(u, v)$, which is the length of the road segment, and each vertex has some traffic lights (see Figure 1). We consider two versions of the *timing setting problem on traffic lights*, or simply the *traffic light* problem: simplified and realistic. In the *simplified traffic light* problem, there is only one traffic light on each vertex. In the *realistic traffic light* problem, the number of traffic lights on each vertex is equal to its degree, and the drivers coming to the vertex from different road segments see different traffic lights, i.e., the traffic coming from each road segment is controlled by a unique traffic light at the vertex. For the simplified version, we use $t(v)$ to denote the starting time of the green light at vertex v , while for the realistic version, we use $t(u, v)$ to denote the starting time of the green light at vertex v which is seen by drivers coming from u .

An example of a traffic network for the simplified traffic light problem is shown in Figure 1. There is a traffic light on every vertex (intersection). In the figure, the number on each vertex represents the starting time of the green light at that vertex. For this simplified traffic light problem, we may simply assume that the vehicles coming from every direction can pass a vertex when the green light is on in the vertex. The realistic traffic light problem gives a more realistic model.

We define a penalty for each edge (v_1, v_2) in E to reflect the goodness of the relative timing settings between $t(v_1)$ and $t(v_2)$ ($t(v_1, v_2)$ and $t(v_2, v_1)$ for the realistic version). We first use an example, as shown in Figure 2(a), to illustrate how to calculate the penalty. In the figure, the road segment (v_1, v_2) has a length of 6 units, which means that it takes 6 units of time for a car to pass through this road segment. Assume that the traffic light

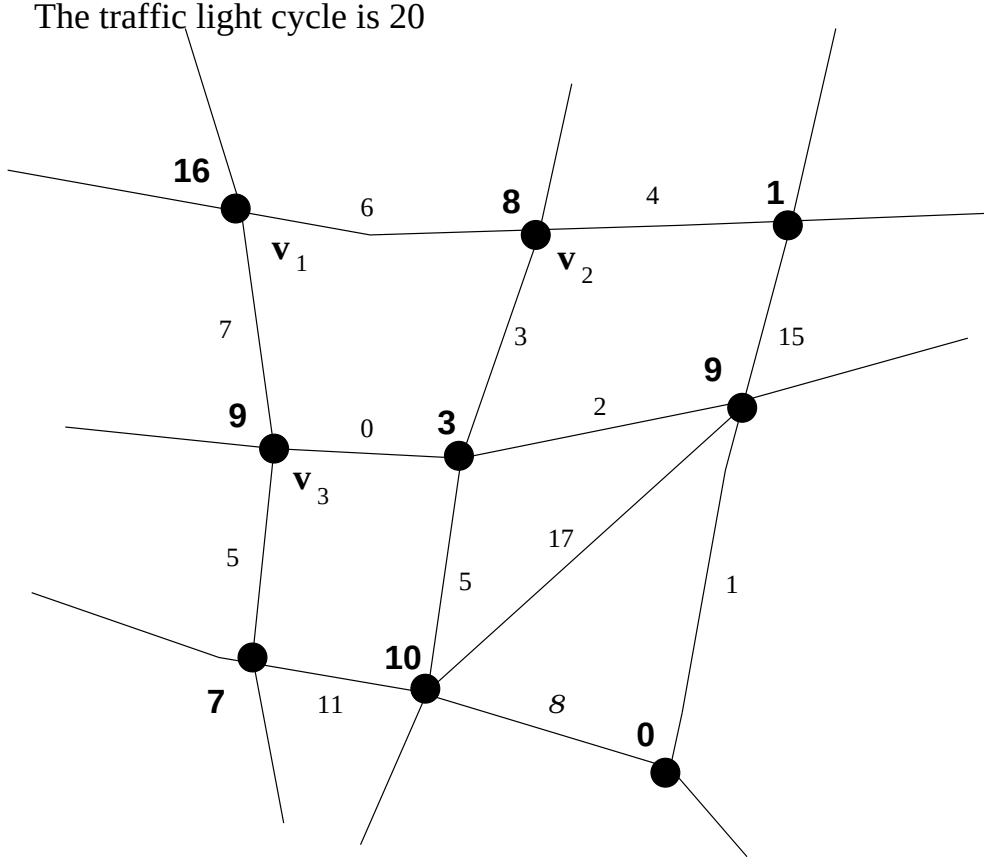


Figure 1: An example of a traffic network .

cycle is 20 units, the red light duration is 10 units, and the green light duration is 10 units. Suppose we set $t(v_1) = 16$ and $t(v_2) = 8$. Figure 2(b) shows the scenario of a car coming from v_1 to v_2 . The thin lines represent the time intervals that the green light is on at vertex v_1 , and hence, a car can pass through v_1 during these intervals. For a car passing v_1 during the green light interval $[-4, 6)$, the dotted line represents the time interval when it arrives at v_2 . The thick line represents the time interval that the green light is on at v_2 . It is easy to see that if a car passes through v_1 during the time interval $[2, 6)$, it will arrive at v_2 at the time interval $[8, 12)$. And since the green light is on at v_2 during this time interval, the car can pass v_2 without waiting. In contrast, if the car passes through v_1 during the time interval $[-4, 2)$, when it arrives at v_2 during the time interval $[2, 8)$, the red light is on at v_2 , and hence, it must wait at v_2 before it can pass v_2 . Therefore, we define the *penalty* from v_1 to v_2 to be the length of the waiting interval $[2, 8)$, which is $8 - 2 = 6$. Similarly, as shown in Figure 2(c), the penalty is defined to be 2 from v_2 to v_1 . The total penalty for the road

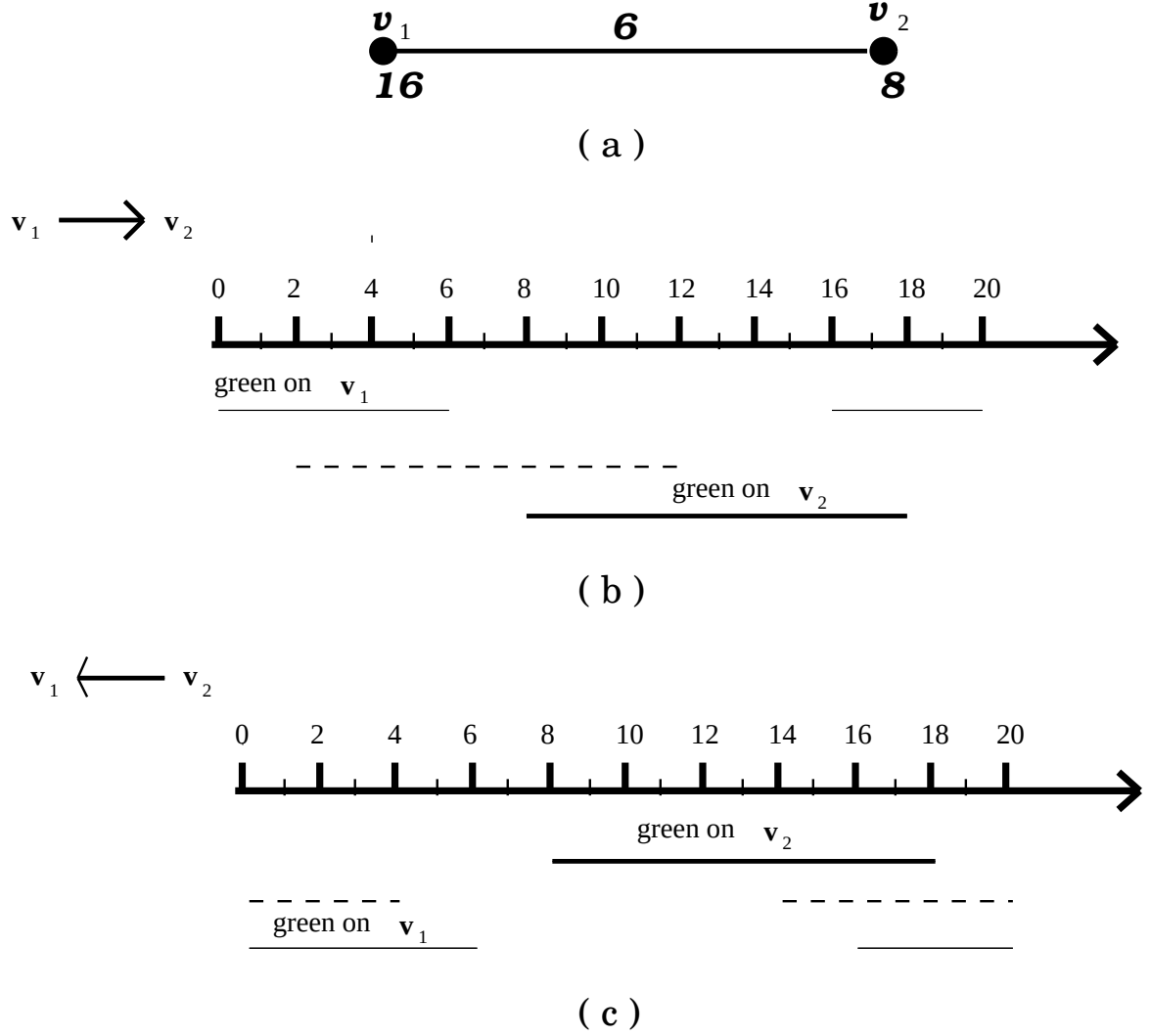


Figure 2: The penalty of an edge with the traffic light cycle 20.(a) The timing setting of an edge. (b)The penalty from v_1 to v_2 . (c)The penalty from v_2 to v_1 .

segment connecting v_1 and v_2 is thus 8. (6 for v_1 to v_2 and 2 for v_2 to v_1).

Let $c_{uv} = |t(v) - d(u, v) - t(u)| \bmod T$ if $(u, v) \in E$ and $c_{uv} = 0$ if $(u, v) \notin E$. It can be easily checked that in general, the penalty from u to v (with respect to the timing settings t) is $P_{uv} = \min\{c_{uv}, T - c_{uv}\}$. Note that P_{uv} may not be equal to P_{vu} . Since each edge has two directions, by definition, the penalty of edge (u, v) is equal to $P_{uv} + P_{vu}$.

Now, we give a formal definition for the *simplified traffic light* problem. Given a traffic light cycle T (T is even) and a graph $G = (V, E)$ in which each edge (u, v) is associated with a nonnegative integral distance $d(u, v)$, we want to assign $t(v)$ for each vertex $v \in V$ such that the total penalty $(\sum_{u \in V} \sum_{v \in V} P_{uv})$ is minimized.

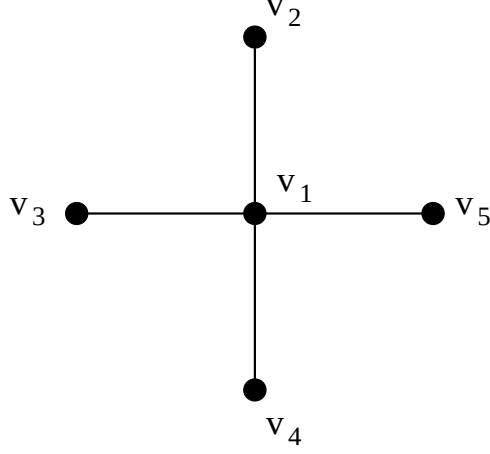


Figure 3: The traffic lights on a vertex v_1 .

For the realistic traffic light problem, the number of traffic lights on each vertex is equal to its degree, and each traffic light on a vertex corresponds to an edge incident to the vertex. Each traffic light on a vertex v is used to control the traffic coming from a road segment (edge) (u, v) . Therefore, we will also use (u, v) to denote the traffic light on vertex v to which the edge (u, v) is incident. We require that the difference between the starting times of the green lights of any two traffic lights on a vertex must be either 0 or $T/2$, and these differences are specified in the problem input. That is, the timing settings for the traffic lights on a vertex are not independent to one another; their relationships are specified in the input. For each traffic light (u, v) , we specify a value $F(u, v)$ in the input, where $F(u, v) = 0$ or 1. The input F values require that $|t(u, v) - t(w, v)| \bmod T = |F(u, v) - F(w, v)| \cdot T/2$, for all $(u, v), (w, v) \in E$. For example, as shown in Figure 3, if we specify in the input that $F(v_2, v_1) = F(v_4, v_1) = 0$ and $F(v_3, v_1) = F(v_5, v_1) = 1$, then when the two south-north direction traffic lights, (v_2, v_1) and (v_4, v_1) , are green, the two east-west direction traffic lights, (v_3, v_1) and (v_5, v_1) , will be red, and vice versa.

The penalties of the edges in the realistic traffic light problem can be defined as follows. Recall that the number $t(u, v)$ is the starting time of the green light of the traffic light on v corresponding to the road segment from u to v . As mentioned above, we assume that $|t(u, v) - t(w, v)| = 0$ or $T/2$. We assume there is no restriction between $t(u, v)$ and $t(v, u)$. Let $C_{uv} = |t(u, v) - d(u, v) - t(v, u)| \bmod T$ if $(u, v) \in E$, and $C_{uv} = 0$ if $(u, v) \notin E$. We define

the penalty from u to v (with respect to the timing settings t) as $P_{uv} = \min\{C_{uv}, T - C_{uv}\}$. The goal of the realistic traffic light problem is then to find assignments $t(u, v)$ and $t(v, u)$ for each edge $(u, v) \in E$ such that the total penalty $(\sum_{u \in V} \sum_{v \in V} P_{uv})$ is minimized.

3 NP-Hardness of the Traffic Light Problem

To prove the NP-hardness of the traffic light problem, we first discuss some related problems. The conventional *vertex coloring* problem [1, 15] is defined as follows. Given a graph $G = (V, E)$, a K -(*vertex*) *coloring* of G is a mapping C from the vertex set V to a set of K colors. A graph G is K -*colorable* if there exists a coloring C with K colors for G such that no two adjacent vertices have the same color, i.e., if $(u, v) \in E$ then $C(u) \neq C(v)$. The vertex coloring problem is to ask, given a graph G and a positive integer K , if G is K -*colorable*. The vertex coloring problem has been proved to be NP-complete for all fixed $K \geq 3$ [8].

We generalize the vertex coloring problem to the following problem.

Problem 1 *Given a graph $G = (V, E)$ and a weight function $w_1 : E \rightarrow \mathbb{R}$, find a K -coloring C that minimizes the following objective function:*

$$W_1(C) = \sum_{(u,v) \in E} \delta_1(u, v) \cdot w_1(u, v),$$

where

$$\delta_1(u, v) = \begin{cases} 1 & \text{if } C(u) = C(v) \\ 0 & \text{otherwise.} \end{cases} \quad \square$$

Note that we allow the weight to be negative. If $w_1(u, v) = 1$ for all $(u, v) \in E$ and we ask if there exists a K -coloring C such that $W_1(C) = 0$, then the problem is equivalent to the conventional vertex coloring problem. This means that the vertex coloring problem is a special case of the unweighted version of Problem 1. Therefore, Problem 1 is also NP-hard for all fixed $K \geq 3$.

Now, consider the following problem.

Problem 2 Given a graph $G = (V, E)$, let $w_2 : E \rightarrow \mathbb{R}^+ \cup \{0\}$ be a weight function and $l : E \rightarrow \{0, 1\}$ a label function on E . The problem is to find a K -coloring C which minimizes the following objective function:

$$W_2(C) = \sum_{(u,v) \in E} \delta_2(u, v) \cdot w_2(u, v),$$

where

$$\delta_2(u, v) = \begin{cases} 1 & \text{if } l(u, v) = 1 \text{ and } C(u) = C(v), \text{ or if } l(u, v) = 0 \text{ and } C(u) \neq C(v) \\ 0 & \text{otherwise.} \end{cases} \quad \square$$

Intuitively, $l(u, v) = 1$ means that u and v should be assigned different colors, otherwise (i.e., $C(u) = C(v)$) we impose a penalty $w_2(u, v)$ on edge (u, v) . Similarly, $l(u, v) = 0$ means that u and v should be assigned the same color, otherwise we impose a penalty $w_2(u, v)$ on the edge.

The following theorem proves that Problem 1 and Problem 2 are, in fact, equivalent.

Theorem 1 Problem 1 and Problem 2 are equivalent.

Proof: Given an instance of Problem 1, we can transform it to an instance of Problem 2 as follows. Set $w_2(u, v) = |w_1(u, v)|$ for all $(u, v) \in E$. If $w_1(u, v) \geq 0$, set $l(u, v) = 1$; otherwise, set $l(u, v) = 0$. Conversely, given an instance of Problem 2, we can transform it to an instance of Problem 1 as follows. If $l(u, v) = 0$, set $w_1(u, v) = -w_2(u, v)$; otherwise, set $w_1(u, v) = w_2(u, v)$.

We have:

$$\begin{aligned} W_1(C) &= \sum_{l(u,v) \in \{0,1\}} \delta_1(u, v) \cdot w_1(u, v) \\ &= \sum_{l(u,v)=1} \delta_1(u, v) \cdot w_1(u, v) + \sum_{l(u,v)=0} \delta_1(u, v) \cdot w_1(u, v) \\ &= \sum_{l(u,v)=1} \delta_1(u, v) \cdot w_2(u, v) - \sum_{l(u,v)=0} \delta_1(u, v) \cdot w_2(u, v) \\ &= \sum_{l(u,v)=1} \delta_2(u, v) \cdot w_2(u, v) - \sum_{l(u,v)=0} \delta_1(u, v) \cdot w_2(u, v) + \sum_{l(u,v)=0} w_2(u, v) \end{aligned}$$

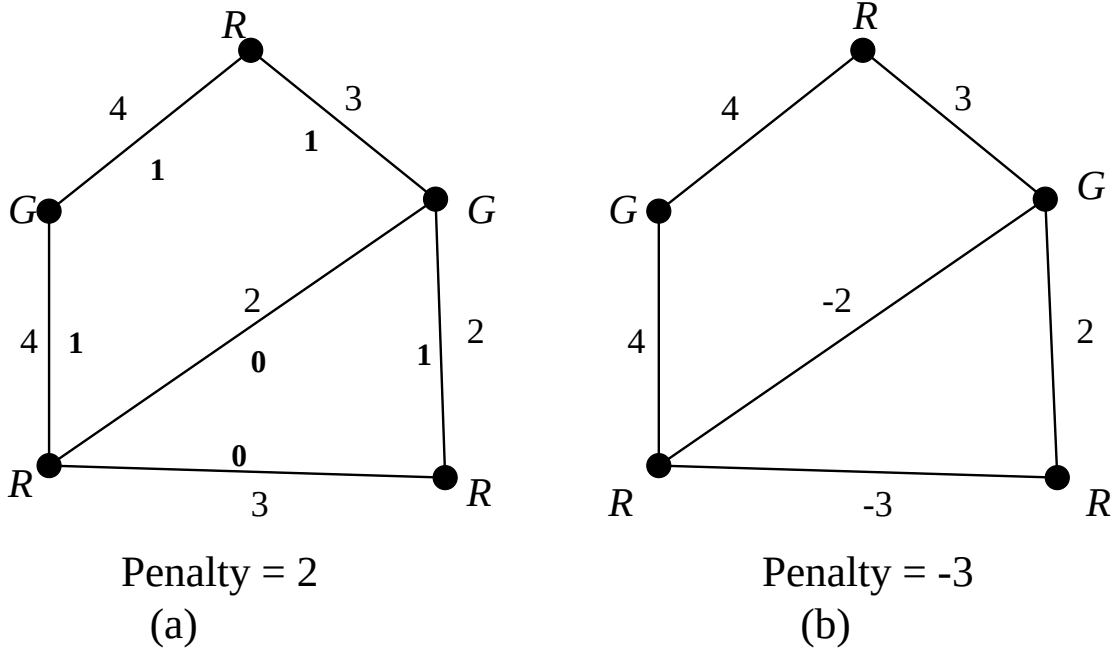


Figure 4: An example for the transformation between of Problem 1 and Problem 2. (a) Problem 2. (b) Problem 1.

$$\begin{aligned}
& - \sum_{l(u,v)=0} w_2(u,v) \\
& = \sum_{l(u,v)=1} \delta_2(u,v) \cdot w_2(u,v) + \sum_{l(u,v)=0} \delta_2(u,v) \cdot w_2(u,v) - \sum_{l(u,v)=0} w_2(u,v) \\
& = \sum_{l(u,v) \in \{0,1\}} \delta_2(u,v) \cdot w_2(u,v) - \sum_{l(u,v)=0} w_2(u,v) \\
& = W_2(C) - \sum_{l(u,v)=0} w_2(u,v) \tag{1}
\end{aligned}$$

Note that $\delta_1(u,v) = \delta_2(u,v)$ if $l(u,v) = 1$, and $\sum_{l(u,v)=0} w_2(u,v) = \sum_{l(u,v)=0} \delta_1(u,v) \cdot w_2(u,v) + \sum_{l(u,v)=0} \delta_2(u,v) \cdot w_2(u,v)$.

Eq. 1 implies minimizing $W_1(C)$ is equivalent to minimizing $W_2(C)$, since given a graph and the weight function $w_2(u,v)$, $\sum_{l(u,v)=0} w_2(u,v)$ is a constant. ■

Figure 4(a) gives an instance of Problem 2. From Theorem 1, the instance of Problem 2 shown in Figure 4(a) can be transformed to an instance of Problem 1 as shown in Figure 4(b). It is easy to see that $W_2(C)$ of Figure 4(a) is 2 and $W_1(C)$ of Figure 4(b) is equal to -3 (note that $\sum_{l(u,v)=0} w_2(u,v) = 5 = W_2(C) - W_1(C)$).

Another graph theory problem related to the traffic light problem is the *maximum cut problem* [1, 14, 15]. A *cut* of a graph $G = (V, E)$ is an edge set $\{(u,v) \mid (u,v) \in E, u \in$

V_1 and $v \in V_2\}$, where V_1 and V_2 is a partition of V , i.e., $V = V_1 \cup V_2$ and $V_1 \cap V_2 = \emptyset$. The maximum cut problem is to find a cut that has the maximum cardinality. If each edge is also associated with a weight, and we want to find a cut that has the maximum total weight, the problem is called the *weighted maximum cut problem*. If we think that all vertices in V_1 are colored with one color, and all vertices in V_2 are colored by another color, then minimizing the total weight of assigning the same color to two adjacent vertices is equivalent to maximizing that of the cut. Thus, the weighted maximum cut problem is equivalent to Problem 1 with $K = 2$.

Now, consider the simplified traffic light problem with $T = 2$. Given a graph $G = (V, E)$ in which each edge (u, v) is associated with a nonnegative integral distance $d(u, v)$, if $d(u, v)$ is odd and $t(u) = t(v)$ then the penalty of (u, v) is $P_{uv} + P_{vu} = \min\{c_{uv}, T - c_{uv}\} + \min\{c_{vu}, T - c_{vu}\} = 2$, where $c_{uv} = |t(v) - d(u, v) - t(u)| \bmod T$. If $d(u, v)$ is odd and $t(v) \neq t(u)$ then the penalty of (u, v) is 0. On the other hand, if $d(u, v)$ is even and $t(v) = t(u)$ then the penalty of (u, v) is 0, and if $d(u, v)$ is even and $t(v) \neq t(u)$ then the penalty of (u, v) is 2. Therefore, if we let $l(u, v) = d(u, v) \bmod T$ and $w_2(u, v) = 2$, it is easy to see that coloring the vertices in V by two colors with finite weight (penalty) is equivalent to assigning $t(v)$ to the vertices. Thus, the simplified traffic light problem with $T = 2$ is equivalent to the unweighted version of Problem 2 with $K = 2$ which, in turn, is equivalent to the unweighted maximum cut problem. Shih *et al.* [14] gave an algorithm of time complexity $O(n^{\frac{3}{2}} \log n)$ to solve the maximum cut problem on a planar graph, where n is the number of vertices in the graph. Therefore, the simplified traffic light problem with $T = 2$ on a planar graph can also be solved in $O((n^{\frac{3}{2}} \log n))$ time.

Moreover, the unweighted version of Problem 2 is a special case of the simplified traffic light problem with $T \geq 3$. Therefore, the simplified traffic light problem with $T \geq 3$ is NP-hard since the unweighted version of Problem 2 is NP-hard.

The realistic traffic light problem on planar graphs with $T = 2$ can also be transformed to Problem 2. Since the number of traffic lights on a vertex is equal to its degree, we will transform each vertex v to a $d(v)$ -polygon, where $d(v)$ is the degree of v . Given a planar

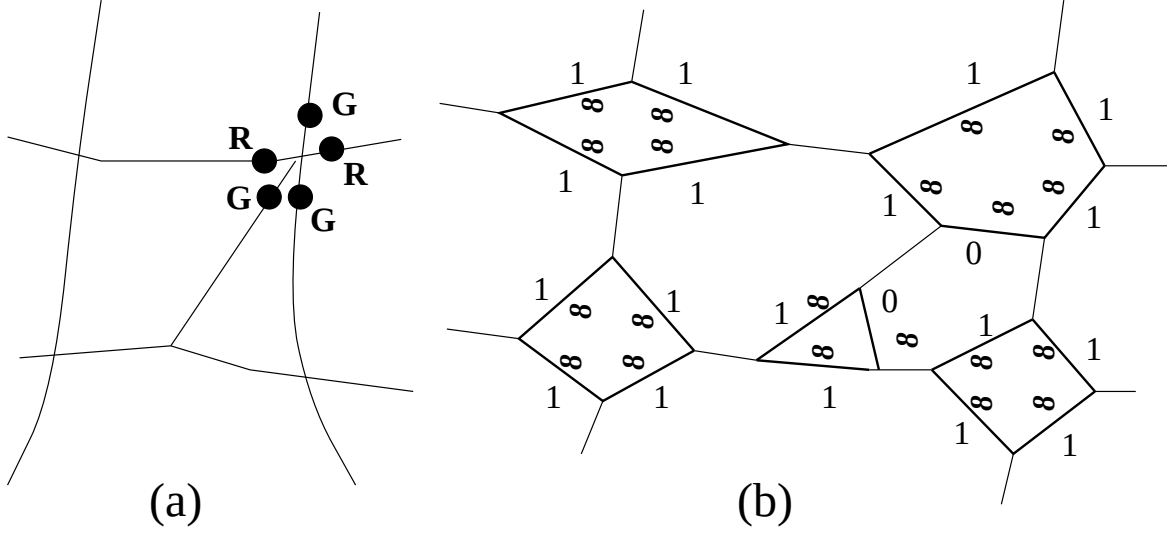


Figure 5: An example of the transformation from the realistic traffic light problem to Problem 2.

graph $G = (V, E)$ with its plane realization in which each edge (u, v) is associated with a nonnegative integral distance $d(u, v)$, we construct a new planar graph $G' = (V', E')$, where $V' = \{v_{ij}, v_{ji} \mid (v_i, v_j) \in E\}$, $E' = E'_1 \cup E'_2$, $E'_1 = \{(v_{ij}, v_{ji}) \mid v_{ij}, v_{ji} \in V'\}$, and $E'_2 = \{(v_{ji}, v_{ki}) \mid v_i, v_j, v_k \in V, (v_j, v_i) \text{ and } (v_k, v_i) \text{ are two neighboring edges in the plane realization of } G\}$. If $(v_{ij}, v_{ji}) \in E'_1$, then set $l(v_{ij}, v_{ji}) = d(v_i, v_j) \bmod T$ and $w_2(v_{ij}, v_{ji}) = 1$. If $(v_{ji}, v_{ki}) \in E'_2$ and we require that $|t(v_{ji}) - t(v_{ki})| \bmod T = 0$, then set $l(v_{ji}, v_{ki}) = 0$ and $w_2(v_{ji}, v_{ki}) = \infty$. If $(v_{ji}, v_{ki}) \in E'_2$ and we require that $|t(v_{ji}) - t(v_{ki})| \bmod T = T/2$, then set $l(v_{ji}, v_{ki}) = 1$ and $w_2(v_{ji}, v_{ki}) = \infty$. It can be shown that a 2-coloring with finite weight (penalty) for the transformed graph corresponds to a valid timing settings for the traffic light problem, and a minimum-weight 2-coloring corresponds to a timing setting for the traffic light problem that has the minimum total penalty. An example for this transformation is shown in Figure 5. Again, using the maximum cut algorithm described in [14], the realistic traffic light problem on planar graphs with $T = 2$ can be solved in time $O(n^{\frac{3}{2}} \log n)$.

4 Properties of the Traffic Light Problem

In this section, we discuss some properties of the simplified traffic light problem. Given a traffic network $G = (V, E)$ and a traffic light cycle T (T is even), we assume, without loss of

generality, that $d(u, v)$ and $t(v)$ are integers, $0 \leq d(u, v) \leq T - 1$, and $0 \leq t(v) \leq T - 1$, for all $v \in V$ and all $(u, v) \in E$ (note that all traffic lights have the same cycle T).

Recall that the penalty (with respect to t) from u to v is defined as $P_{uv} = \min\{c_{uv}, T - c_{uv}\}$, where $c_{uv} = |t(v) - d(u, v) - t(u)| \bmod T$. For convenience of discussion, we define the *penalty function* f_T as follows.

- $f_T(x + T) = f_T(x)$;
- $f_T(x) = x$, if $0 \leq x \leq T/2$;
- $f_T(x) = T - x$, if $T/2 < x \leq T$.

Using the penalty function f_T , the penalty P_{uv} from u to v can now be denoted as $f_T(t(v) - d(u, v) - t(u))$. For example, in Figure 2(a), the penalty from v_1 to v_2 is $P_{12} = f_{20}(8 - 6 - 16) = f_{20}(-14) = f_{20}(6) = 6$ and the penalty from v_2 to v_1 is $P_{21} = f_{20}(16 - 6 - 8) = 2$. Thus, the penalty of edge (v_1, v_2) is equal to $P_{12} + P_{21} = 8$.

We define the *extra penalty* of an edge as the actual penalty under a certain setting minus the minimum penalty of the edge. For example, in Figure 2(a), if we set $t(v_1) = 16$ and $t(v_2) = 11$, then the penalty of (v_1, v_2) is equal to $P_{12} + P_{21} = f_{20}(11 - 6 - 16) + f_{20}(16 - 6 - 11) = 9 + 1 = 10$. However, the minimum penalty of (v_1, v_2) is equal to 8 (A setting method will be given later.). Thus, the extra penalty of this edge under this setting is $10 - 8 = 2$.

We now prove some properties of the penalty function f_T in the following theorem.

Theorem 2 (1) $f_T(f_T(x)) = f_T(x)$.

(2) $f_T(-x) = f_T(x)$.

(3) $f_T(x + T/2) = f_T(x - T/2) = T/2 - f_T(x)$.

(4) $f_T(x + y) \leq f_T(x) + f_T(y)$.

Proof: (1) Assume $f_T(x) = k$. By definition, we have $0 \leq k \leq T/2$, and $f_T(f_T(x)) = f_T(k) = k$. Therefore, $f_T(f_T(x)) = f_T(x)$.

(2) Let $x = n \cdot T + k$, where $n \in \mathbb{Z}$ and $0 \leq k < T$. We have $f_T(x) = f_T(n \cdot T + k) = f_T(k)$ and

$f_T(-x) = f_T(-k)$. If $0 \leq k \leq T/2$, $f_T(x) = k$ and $f_T(-k) = f_T(-k+T) = T - (-k+T) = k$. If $T/2 < k < T$, $f_T(x) = T - k$ and $f_T(-k) = f_T(-k+T) = T - k$. Therefore, $f_T(-x) = f_T(x)$.

(3) $f_T(x - T/2) = f_T(x - T/2 + T) = f_T(x + T/2)$. Let $x = n \cdot T + k$, where $n \in \mathbb{Z}$ and $0 \leq k < T$. If $0 \leq k \leq T/2$, $f_T(x + T/2) = f_T(k + T/2) = T/2 - k = T/2 - f_T(x)$. If $T/2 < k < T$, $f_T(x + T/2) = f_T(k + T/2) = f_T(k - T/2) = k - T/2$ and $T/2 - f_T(x) = T/2 - (T - k) = k - T/2$. Therefore, $f_T(x + T/2) = f_T(x - T/2) = T/2 - f_T(x)$.

(4) Let $x = n \cdot T + k$ and $y = m \cdot T + l$, where $n, m \in \mathbb{Z}$ and $0 \leq k, l < T$. We need to consider the following six cases:

Case 1: If $0 \leq k, l \leq T/2$ and $0 \leq (k+l) \leq T/2$, $f_T(x+y) = f_T(k+l) = k+l = f_T(x) + f_T(y)$.

Case 2: If $0 \leq k, l \leq T/2$ and $T/2 < (k+l) \leq T$, $f_T(x+y) = f_T(k+l) = T - (k+l) < (k+l) = f_T(x) + f_T(y)$.

Case 3: If $0 \leq k \leq T/2$ and $T/2 < l \leq T$ and $T/2 < (k+l) \leq T$, $f_T(x+y) = f_T(k+l) = T - (k+l) < k + T - l = f_T(x) + f_T(y)$.

Case 4: If $0 \leq k \leq T/2$ and $T/2 < l \leq T$ and $T < (k+l) \leq 3T/2$, $f_T(x+y) = f_T(k+l) = (k+l) - T < k + T - l = f_T(x) + f_T(y)$.

Case 5: If $T/2 < k, l < T$ and $T < (k+l) \leq 3T/2$, $f_T(x+y) = f_T(k+l) = k+l - T < T - (k+l - T) = f_T(x) + f_T(y)$.

Case 6: If $T/2 < k, l < T$ and $3T/2 < (k+l) \leq 2T$, $f_T(x+y) = f_T(k+l) = 2T - (k+l) = T - k + T - l = f_T(x) + f_T(y)$.

From Cases 1 through 6, we conclude that $f_T(x+y) \leq f_T(x) + f_T(y)$. ■

Next, we consider a special kind of timing setting method which requires that the penalties of both directions of every edge are the same, that is, $P_{uv} = P_{vu}$ for each $(u, v) \in E$. For $(v_1, v_2) \in E$, given a traffic light cycle T , if we set $t(v_1) = t(v_2) = k$, $0 \leq k < T$, then $P_{12} = f_T(t(v_2) - d(v_1, v_2) - t(v_1)) = f_T(k - d(v_1, v_2) - k) = f_T(d(v_1, v_2)) = P_{21}$. And if we set $t(v_1) = k$ and $t(v_2) = (k + T/2) \bmod T$, $0 \leq k < T$, then $P_{12} = f_T(k + T/2 - d(v_1, v_2) - k) = f_T(d(v_1, v_2) - T/2)$ (by Theorem 2), and $P_{21} = f_T(k - d(v_1, v_2) - k - T/2) = f_T(d(v_1, v_2) + T/2)$, thus $P_{12} = P_{21}$ (by Theorem 2). Hence, if we set $t(v_1) = k$, to get the same penalties for

both directions of (v_1, v_2) , it can be easily shown that we have only two choices to set $t(v_2)$: k or $(k + T/2) \bmod T$. We call this simplified traffic light problem with the above timing setting restriction the *two-phase setting problem*, since there are only two possible “phases,” $k \bmod T$ and $|k + T/2| \bmod T$, for the timing settings of all vertices in V . Note that the total penalty is independent of the actual value of k .

Tables 1 and 2 list the assignments for $t(v_2)$ that will give the smaller and the larger values of $P_{12} + P_{21} = 2P_{12}$, respectively.

Table 1 *The smaller penalty on an edge (v_1, v_2) .*

condition	$t(v_1)$	$t(v_2)$	smaller penalty
$0 \leq d(v_1, v_2) \leq \frac{T}{4}$	k	k	$2d(v_1, v_2)$
$\frac{T}{4} < d(v_1, v_2) \leq \frac{T}{2}$	k	$k + \frac{T}{2}$	$T - 2d(v_1, v_2)$
$\frac{T}{2} < d(v_1, v_2) \leq \frac{3T}{4}$	k	$k + \frac{T}{2}$	$2d(v_1, v_2) - T$
$\frac{3T}{4} < d(v_1, v_2) \leq T$	k	k	$2T - 2d(v_1, v_2)$

Table 2 *The larger penalty on an edge (v_1, v_2) .*

condition	$t(v_1)$	$t(v_2)$	larger penalty
$0 \leq d(v_1, v_2) \leq \frac{T}{4}$	k	$k + \frac{T}{2}$	$T - 2d(v_1, v_2)$
$\frac{T}{4} < d(v_1, v_2) \leq \frac{T}{2}$	k	k	$2d(v_1, v_2)$
$\frac{T}{2} < d(v_1, v_2) \leq \frac{3T}{4}$	k	k	$2T - 2d(v_1, v_2)$
$\frac{3T}{4} < d(v_1, v_2) \leq T$	k	$k + \frac{T}{2}$	$2d(v_1, v_2) - T$

It is easy to check that the following transformation from the two-phase setting problem to Problem 2 with $K = 2$ is valid. Let $l(v_1, v_2) = 1$ if $T/4 < d(v_1, v_2) \leq 3T/4$ and $l(v_1, v_2) = 0$ otherwise. Let $w_2(v_1, v_2) = 2 \cdot |f_T(t(k) - d(v_1, v_2) - t(k)) - f_T(t(k) - d(v_1, v_2) - t(k + T/2))|$. Therefore, again, we can solve the two-phase setting problem on planar graphs in $O(n^{\frac{3}{2}} \log n)$ time by the algorithm designed for the weighted maximum cut problem.

Note, however, that we may not always get the minimum total penalty using the two-phase setting method. We give an example to illustrate the non-optimality of the two-phase setting method as follow.

Example 1 *Consider the graph shown in Figure 6. Assume the traffic light cycle $T = 20$. If we set each vertex to either k or $k + \frac{T}{2}$, then the penalty is 24 in the assignment: $t(v_0) = 0$, $t(v_1) = 0$, $t(v_2) = T/2$ and $t(v_3) = T/2$. And 24 is the minimum for the two-phase setting problem. However, if we do not require that the penalties of both directions of an edge are*

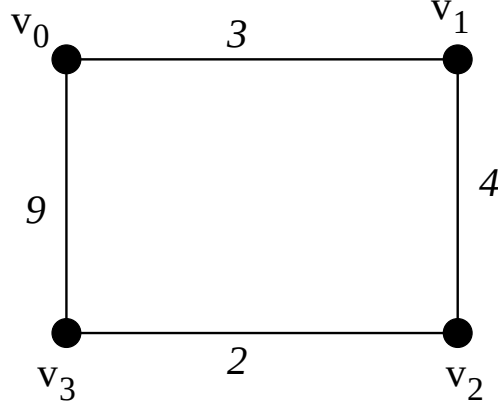


Figure 6: A simple traffic network.

the same, we can obtain the minimum penalty 20 with the following assignment: $t(v_0) = 7$, $t(v_1) = 4$, $t(v_2) = 0$ and $t(v_3) = 18$. Thus, if we only assign each vertex to either k or $k + \frac{T}{2}$, it does not guarantee to find the minimum penalty of a graph.

The following lemma gives a method to find the minimum penalty of an edge.

Lemma 1 *Given an edge (v_1, v_2) with distance $d(v_1, v_2)$, the minimum penalty of (v_1, v_2) is equal to $f_T(2d(v_1, v_2))$ and the maximum penalty of (v_1, v_2) is equal to $T - f_T(2d(v_1, v_2))$. Besides, the minimum penalty can be obtained if the penalty of one direction of the edge is set to zero.*

Proof: Let P_{12} and P_{21} denote the penalty from v_1 to v_2 and from v_2 to v_1 , respectively. The penalty of (v_1, v_2) is $P_{12} + P_{21} = f_T(t(v_2) - d(v_1, v_2) - t(v_1)) + f_T(t(v_1) - d(v_1, v_2) - t(v_2))$. Because the penalty function is a period function, as shown in Figure 7, it is clear that $P_{12} + P_{21}$ has the minimum value when $P_{12} = 0$ or $P_{21} = 0$. Suppose $t(v_1) = k$. $P_{12} = 0$ if $t(v_2)$ is set to $d(v_1, v_2) + k$. Thus, the minimum penalty of an edge (v_1, v_2) is equal to $f_T((d(v_1, v_2) + k) - d(v_1, v_2) - k) + f_T(k - d(v_1, v_2) - (d(v_1, v_2) + k)) = f_T(2d(v_1, v_2))$.

It is also clear that $P_{12} + P_{21}$ has the maximum value when $P_{12} = T/2$ or $P_{21} = T/2$ (See Figure 7). Suppose $t(v_1) = k$. Similarly, $P_{12} = T/2$ if $t(v_2)$ set to $T/2 + d(v_1, v_2) + k$. The maximum penalty of the edge is equal to $f_T((d(v_1, v_2) + k + T/2) - d(v_1, v_2) - k) + f_T(k - d(v_1, v_2) - (d(v_1, v_2) + k + T/2)) = f_T(\frac{T}{2}) + f_T(-\frac{T}{2} - 2d(v_1, v_2)) = \frac{T}{2} + \frac{T}{2} - f_T(-2d(v_1, v_2)) = T - f_T(2d(v_1, v_2))$. ■

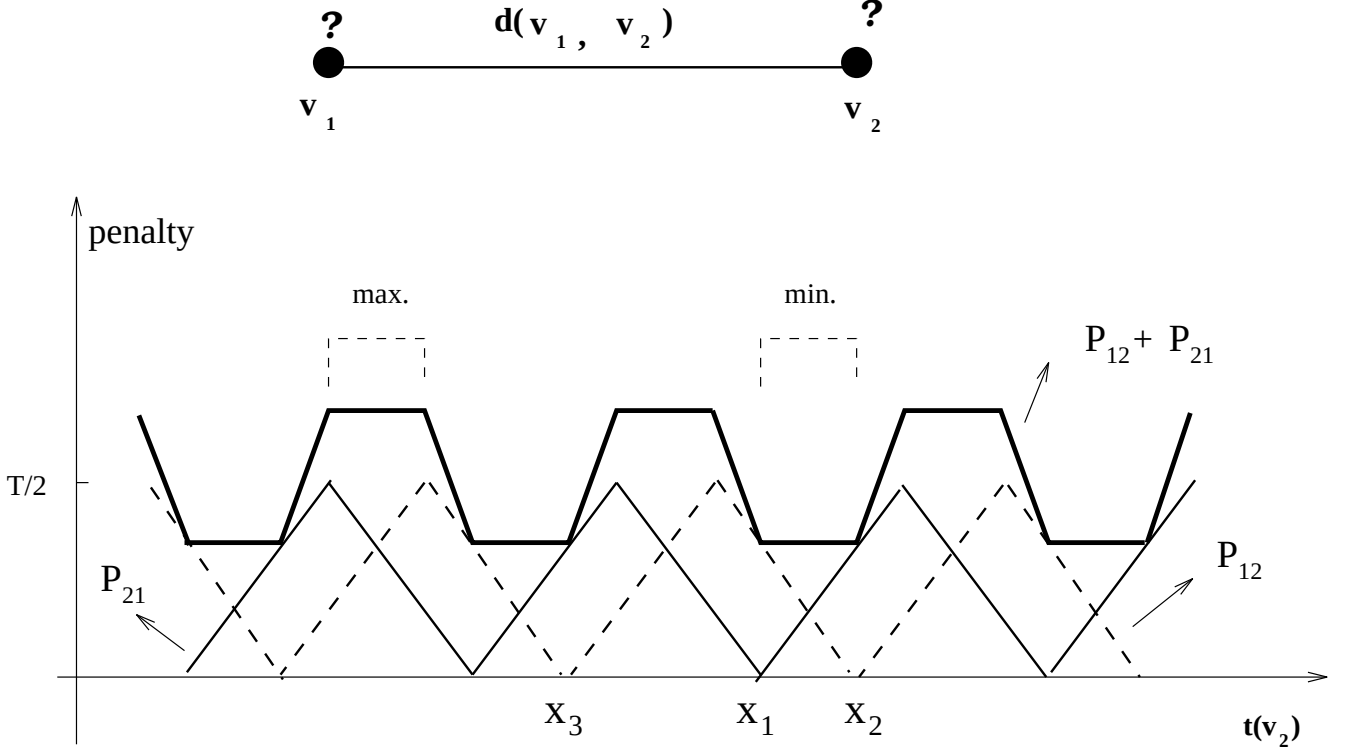


Figure 7: The penalty of an edge.

Lemma 2 Given a graph $G = (V, E)$, where $V = \{v_i \mid 1 \leq i \leq n+1\}$ and $E = \{(v_i, v_{i+1}) \mid 1 \leq i \leq n\}$ (G is a path of $n+1$ vertices), the minimum penalty of G can be obtained by setting either $P_{ij} = 0$ or $P_{ji} = 0$ for every edge and the minimum penalty of G is equal to $\sum_{i=1}^n f_T(2d(v_i, v_{i+1}))$ and the maximum penalty of G is equal to $n \cdot T - \sum_{i=2}^n f_T(2d(v_i, v_{i+1}))$.

Proof: If the penalty of every edge is the minimum, then we will get the minimum penalty of G . We set $t(v_1) = 0$ and $t(v_{i+1}) = (t(v_i) + d(v_i, v_{i+1})) \bmod T$, $1 \leq i \leq n$, then the penalty of every edge will be the minimum. Thus, by Lemma 1, the minimum penalty of G is $\sum_{i=1}^n f_T(2d(v_i, v_{i+1}))$.

Similarly, we can get the maximum penalty of a path if the penalty of each edge is maximized. Thus, by Lemma 1, the maximum penalty of G is $\sum_{i=1}^n (T - f_T(2d(v_i, v_{i+1}))) = n \cdot T - \sum_{i=1}^n f_T(2d(v_i, v_{i+1}))$. ■

In fact, the minimum penalty of an edge (u, v) does not occur only at $P_{uv} = 0$ or $P_{vu} = 0$. Lemma 3 shows the solution is a range dependent on $d(u, v)$.

Lemma 3 *Given an edge (v_1, v_2) with distance $d(v_1, v_2)$, if $t(v_1)$ has been set, then we will get the minimum penalty of (v_1, v_2) when $t(v_2)$ is any integer number between $f_T(t(v_1) - d(v_1, v_2))$ and $f_T(t(v_1) + d(v_1, v_2))$. In other words, the number of solutions of $t(v_2)$ for minimizing the penalty of (v_1, v_2) is $f_T(2d(v_1, v_2)) + 1$.*

Proof: See Figure 7. We have that $P_{21} = 0$ if $t(v_2) = x_1 = (t(v_1) - d(v_1, v_2)) \bmod T + mT$ and $P_{12} = 0$ if $t(v_2) = x_2 = (t(v_1) + d(v_1, v_2)) \bmod T + nT$ or $t(v_2) = x_3 = (t(v_1) + d(v_1, v_2)) \bmod T + (n - 1)T$, where m and n are integers, $x_2 = x_3 + T$ and $x_3 < x_1 < x_2$. The range for minimizing the penalty of (v_1, v_2) is either $[x_3, x_1]$ or $[x_1, x_2]$. Let r be the number of solutions of $t(v_2)$ for minimizing the penalty of (v_1, v_2) . We have $r \leq T - r$. Thus, $r = \min(x_3 - x_1 + 1, x_2 - x_1 + 1) = \min((-2d(v_1, v_2) \bmod T) + 1, (2d(v_1, v_2) \bmod T) + 1) = f_T(2d(v_1, v_2)) + 1$. Thus, the number of solutions of $t(v_2)$ is $f_T(2d(v_1, v_2)) + 1$. ■

Theorem 3 will give a method to find the minimum penalty of two adjacent edges when the two endpoints have been set (See Figure 8(a)).

Theorem 3 *Given two adjacent edges (v_1, v_2) and (v_2, v_3) , suppose $t(v_1)$ and $t(v_3)$ have been set, and $f_T(2d(v_1, v_2)) \leq f_T(2d(v_2, v_3))$. The minimum penalty for these two edges can be found if we try to assign each integer between $f_T(t(v_1) - d(v_1, v_2))$ and $f_T(t(v_1) + d(v_1, v_2))$ to $t(v_2)$. Besides, the setting of $t(v_2)$ for either $P_{12} = 0$, i.e. $t(v_2) = f_T(t(v_1) + d(v_1, v_2))$, or $P_{21} = 0$, i.e. $t(v_2) = f_T(t(v_1) - d(v_1, v_2))$, is in the range. The maximum penalty for these two edges can be found if we try to assign each integer between $f_T(t(v_1) - d(v_1, v_2) + T/2)$ and $f_T(t(v_1) + d(v_1, v_2) + T/2)$ to $t(v_2)$. Besides, the setting of $t(v_2)$ for either $P_{12} = T/2$, i.e. $t(v_2) = f_T(t(v_1) + d(v_1, v_2) + T/2)$, or $P_{21} = T/2$ i.e., $t(v_2) = f_T(t(v_1) - d(v_1, v_2) + T/2)$, is in the range. And, the maximum penalty plus the minimum penalty is equal to $2T$.*

Proof: Because $f_T(2d(v_1, v_2)) \leq f_T(2d(v_2, v_3))$, by Lemma 1, the minimum value of $P_{12} + P_{21}$ is no more than $P_{23} + P_{32}$. Suppose $P_{12} = 0$, $P_{21} = 0$, $P_{23} = 0$ and $P_{32} = 0$ occur at $t(v_2) = a$, $t(v_2) = b$, $t(v_2) = c$ and $t(v_2) = d$ respectively. We have to consider the following cases.

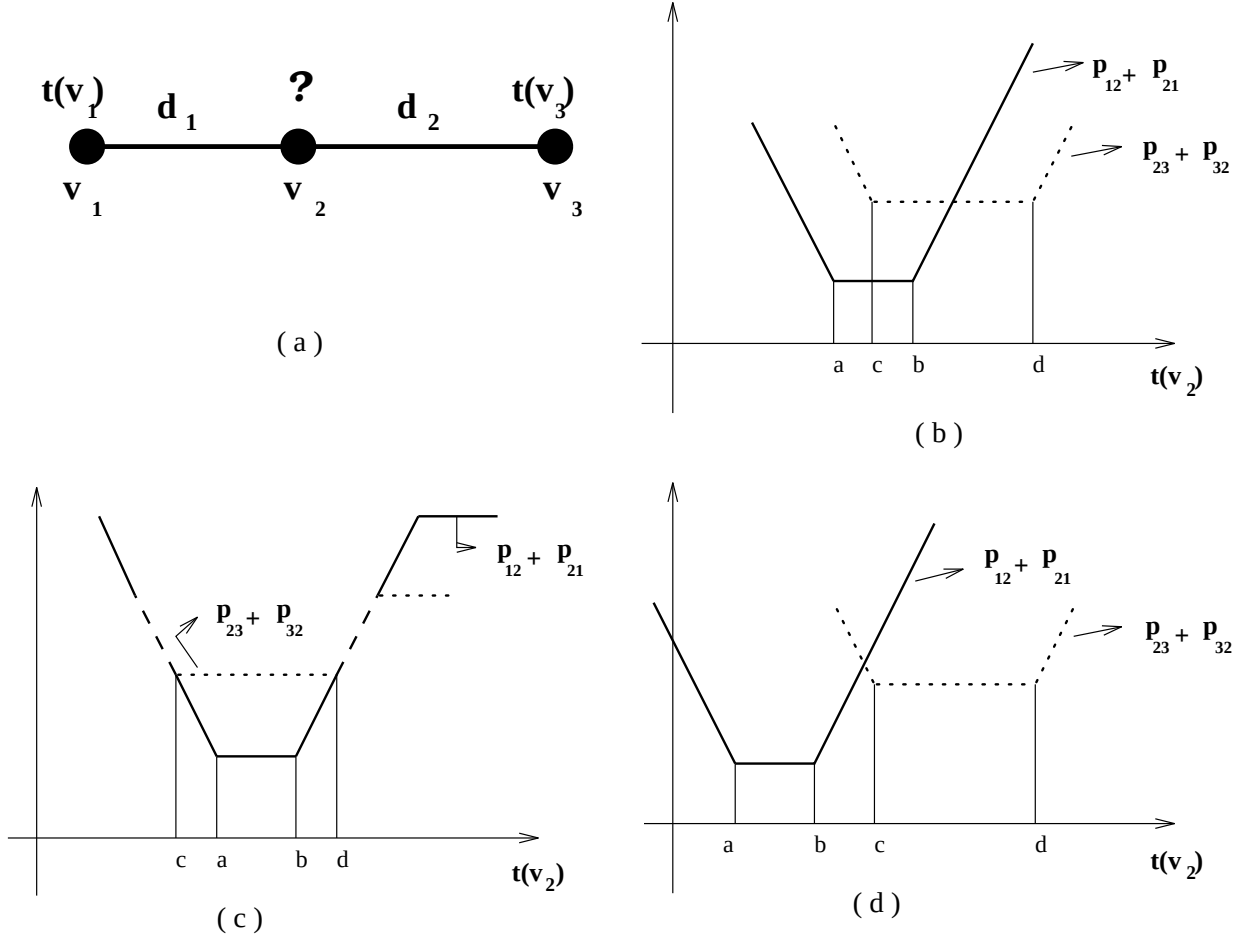


Figure 8: Three cases for the penalty of two adjacent edges.

Case 1: $a < c < b < d$ or $c < a < d < b$. See Figure 8(b). The minimum penalty of edges (v_1, v_2) and (v_2, v_3) occurs in $[c, b]$ or $[a, d]$. And $[c, b]$ or $[a, d]$ is contained in the range between $f_T(t(v_1) - d(v_1, v_2))$ and $f_T(t(v_1) + d(v_1, v_2))$. Besides, the setting of $t(v_2)$ for either $P_{21} = 0$ or $P_{12} = 0$ (b or a) is in the range.

Case 2: $c < a < b < d$. See Figure 8(c). The minimum penalty occurs in $[a, b]$. And $[a, b]$ is contained in the range between $f_T(t(v_1) - d(v_1, v_2))$ and $f_T(t(v_1) + d(v_1, v_2))$. Besides, the setting of $t(v_2)$ for $P_{12} = 0$ or $P_{21} = 0$ (a or b) is in the range.

Case 3: $a < b < c < d$ or $c < d < a < b$. See Figure 8(d). The minimum penalty will occur in $[b, c]$ or $[d, a]$. And b or a is contained in the range between $f_T(t(v_1) - d(v_1, v_2))$ and $f_T(t(v_1) + d(v_1, v_2))$. Besides, the setting of $t(v_2)$ for either $P_{21} = 0$ or $P_{12} = 0$ (b or a) is in the range.

By cases 1, 2 and 3, we get the range for setting the minimum penalty of the two adjacent edges. It is also obvious that the point at either $P_{12} = 0$ or $P_{21} = 0$ is in the range.

By the similar proof technique, we can prove that the case for the maximum penalty is also correct.

Now, we can find the minimum penalty of the two adjacent edges for testing the two possible values for $t(v_2)$: $t(v_2)=t(v_1) + d(v_1, v_2)$ ($P_{12} = 0$) or $t(v_2)= t(v_1) - d(v_1, v_2)$ ($P_{21} = 0$). If $t(v_2)=t(v_1) + d(v_1, v_2)$, the minimum penalty is $P_a = f_T(2d(v_1, v_2)) + f_T(t(v_1) + d(v_1, v_2) - d(v_2, v_3) - t(v_3)) + f_T(t(v_3) - d(v_2, v_3) - t(v_1) - d(v_1, v_2))$. If $t(v_2)=t(v_1) - d(v_1, v_2)$, the minimum penalty is $P_b = f_T(2d(v_1, v_2)) + f_T(t(v_1) - d(v_1, v_2) - d(v_2, v_3) - t(v_3)) + f_T(t(v_3) - d(v_2, v_3) - t(v_1) + d(v_1, v_2))$. Thus, the minimum penalty of these two edges is the minimum of P_a and P_b .

Now, we can also find the maximum penalty of the two adjacent edges for testing the two possible values for $t(v_2)$: $t(v_2)=t(v_1) + d(v_1, v_2) + T/2$ ($P_{12} = T/2$) or $t(v_2)= t(v_1) - d(v_1, v_2) + T/2$ ($P_{21} = T/2$). If $t(v_2)=t(v_1) + d(v_1, v_2) + T/2$, the maximum penalty is $P_c = T - f_T(2d(v_1, v_2)) + f_T(t(v_1) + d(v_1, v_2) + T/2 - d(v_2, v_3) - t(v_3)) + f_T(t(v_3) - d(v_2, v_3) - t(v_1) - d(v_1, v_2) - T/2)$. By Theorem 2(3), $P_c = 2T - P_a$. If $t(v_2)=t(v_1) - d(v_1, v_2) + T/2$, the maximum penalty is $P_d = T - f_T(2d(v_1, v_2)) + f_T(t(v_1) - d(v_1, v_2) + T/2 - d(v_2, v_3) - t(v_3)) + f_T(t(v_3) - d(v_2, v_3) - t(v_1) + d(v_1, v_2) - T/2)$. By Theorem 2(3), $P_d = 2T - P_b$. The maximum penalty of these two edges is the maximum of P_c and P_d . Thus, the maximum penalty of these two edges is $2T -$ (the minimum penalty). ■

We extend Theorem 3 to a graph consisting of a path with n edges.

Theorem 4 *Given a graph $G = (V, E)$, where $V = \{v_i \mid 1 \leq i \leq n+1\}$, $E = \{ (v_i, v_{i+1}) \mid 1 \leq i \leq n \}$ and $d_i = d(v_i, v_{i+1})$, $1 \leq i \leq n$, under the condition that $t(v_1)$ and $t(v_{n+1})$ have been set, if $f_T(2d_n) \geq \max\{f_T(2d_1), f_T(2d_2), \dots, f_T(2d_{n-1})\}$, then the minimum penalty of G can be found by testing the solution set S , where S consists of all possible settings such that (v_1, v_2) , (v_2, v_3) , $\dots$, (v_{n-2}, v_{n-1}) and (v_{n-1}, v_n) have zero extra penalty.*

Proof: We shall prove this theorem by induction on n . At the basis, by Theorem 3, it is true when $n = 2$. Suppose $n < k$ is true. For $n = k$, assume (v_1, v_2) has the minimum penalty when $t(v_2)$ is any integer in $[a, b]$, under the condition that $t(v_1)$ has been set. For

a graph $G' = (V', E')$, where $V' = \{v_i \mid 2 \leq i \leq k+1\}$ and $E' = \{(v_i, v_{i+1}) \mid 2 \leq i \leq k\}$, because G' has $k-1$ edges, we can find the minimum penalty of G' by testing the solution set S' , where S' consists of all possible settings such that $(v_2, v_3), (v_3, v_4), \dots, (v_{k-2}, v_{k-1})$ and (v_{k-1}, v_k) have zero extra penalty. Let P' be the minimum penalty of G' when we set $t(v_2) = x_2$, where $a \leq x_2 \leq b$ and $t(v_i) = x_i$, where $3 \leq i \leq k$. And, suppose, under this setting, the penalty of (v_i, v_{i+1}) is P_i , $1 \leq i \leq k-1$. Thus, the penalty of G is $P_1 + P'$ and only (v_k, v_{k+1}) is not guaranteed to have zero extra penalty.

Assume we change $t(v_2)$ to y where y is not in $[a, b]$ and $t(v_i)$ to $(x_i + y - x_2) \bmod T$, $3 \leq i \leq k$. The penalties of all edges in G are not changed except (v_1, v_2) and (v_k, v_{k+1}) . Suppose the penalties of (v_1, v_2) and (v_k, v_{k+1}) are changed to $P_1 + z_1$ and $P_k - z_2$ respectively. Because $f_T(2d(v_1, v_2)) \leq f_T(2d(v_k, v_{k+1}))$, we have $z_1 \geq z_2$. Thus, the penalty of G becomes $P_1 + P' + z_1 - z_2$, which is not smaller than $P_1 + P'$, when $t(v_2) = y$, $y \notin [a, b]$ and $t(v_i) = (x_i + y - x_2) \bmod T$, where $3 \leq i \leq k$. This completes the proof. ■

In Theorem 4, if we combine the starting vertex and the ending vertex of a path into a single vertex, we will get a polygon. Thus, we have a similar property for a polygon as follows.

Corollary 1 *Given a graph $G = (V, E)$, where $V = \{v_i \mid 1 \leq i \leq n\}$, $E = \{(v_1, v_2), (v_2, v_3), \dots, (v_{n-1}, v_n), (v_n, v_1)\}$, $d_i = d(v_i, v_{i+1})$, $1 \leq i \leq n-1$ and $d_n = d(v_n, v_1)$, if $f_T(2d_n) \geq \max\{f_T(2d_1), f_T(2d_2), \dots, f_T(2d_{n-1})\}$, then the minimum penalty of G can be found by testing the solution set S , where S consists of all possible settings such that $(v_1, v_2), (v_2, v_3), \dots, (v_{n-2}, v_{n-1})$ and (v_{n-1}, v_n) have zero extra penalty.*

For the realistic traffic light problem, the number of traffic lights on each vertex is equal to its degree. In this problem, we assume the difference of the starting times of the green lights of different traffic lights on the same vertex is either 0 or $\frac{T}{2}$. We give an example to illustrate how to transform the realistic traffic light problem on a polygon to the simplified traffic light problem.

Example 2 *Consider the triangle shown in Figure 9(a). Assume that $f_T(2d_3) \geq \max\{$*

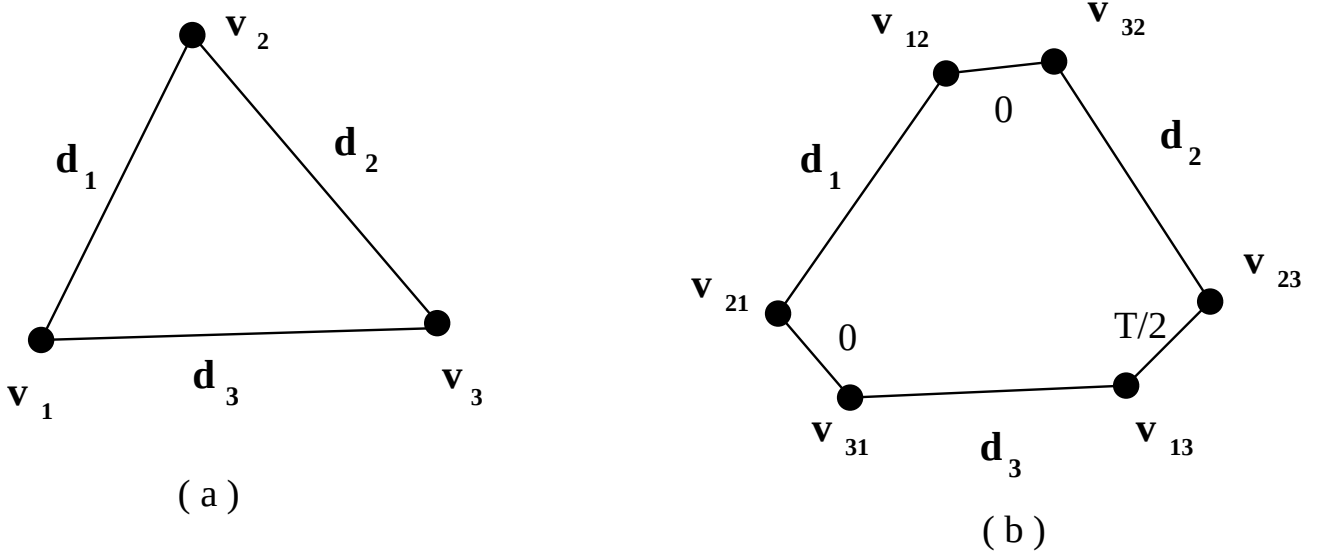


Figure 9: An example for the realistic traffic light problem (a) A triangle. (b) The graph after transformation.

$f_T(2d_2), f_T(2d_1)\}$. In Figure 9(a), each vertex has two traffic lights. Assume the traffic lights on v_1 have the same color at the same time, the traffic lights on v_2 have the same color at the same time, and the traffic lights on v_3 have different colors at the same time. For calculating the minimum penalty of the triangle, we transform each vertex to an edge. After the transformation, each vertex represents a traffic light, as shown in Figure 9(b). Because the traffic lights on v_1 and v_2 have the same color at the same time respectively, we set $d(v_{21}, v_{31}) = 0$ and $d(v_{12}, v_{32}) = 0$. Because the traffic lights on v_3 have different colors at the same time, we set $d(v_{13}, v_{23}) = T/2$. Since each vertex on this hexagon has only one traffic light, by Corollary 1, we can find the minimum penalty of this hexagon by testing the solution set in which each setting makes (v_{31}, v_{21}) , (v_{21}, v_{12}) , (v_{12}, v_{32}) , (v_{32}, v_{23}) and (v_{23}, v_{13}) have zero extra penalty.

Theorem 5 shows that the realistic traffic light problem on a polygon can be reduced to the simplified traffic light problem.

Theorem 5 Given a graph $G = (V, E)$, where $V = \{v_i | 1 \leq i \leq n\}$, $E = \{(v_1, v_2), (v_2, v_3), \dots, (v_{n-1}, v_n), (v_n, v_1)\}$, $d_i = d(v_i, v_{i+1})$, $1 \leq i \leq n-1$ and $d_n = d(v_n, v_1)$, the realistic traffic light problem can be reduced to the simplified traffic light problem.

Proof: For convenient representation, let $v_{n+1} = v_1$. For each vertex v_i , we transform it to an edge $(v_{(i-1)i}, v_{(i+1)i})$. If the two traffic lights on v_i must have the same color at the same time, i.e. $|t(v_{i-1}, v_i) - t(v_{i+1}, v_i)| = 0$, then we set $d(v_{(i-1)i}, v_{(i+1)i}) = 0$ and if otherwise, i.e. $|t(v_{i-1}, v_i) - t(v_{i+1}, v_i)| = \frac{T}{2}$, set $d(v_{(i-1)i}, v_{(i+1)i}) = \frac{T}{2}$. Let G' denote the new graph. It is clear that each vertex in G' has one traffic light. Because the minimum penalty of each new edge is 0 ($f_T(2 \cdot \frac{T}{2}) = 0$ and $f_T(2 \cdot 0) = 0$), the penalty of each new edge will be 0 when we find the minimum penalty of G' by Corollary 1. Thus, the minimum penalty of G' for the simplified traffic light problem is equal to the minimum penalty of G for the realistic traffic light problem. In other words, the realistic traffic light problem on a graph consisting of a polygon is reduced to the simplified traffic light problem. ■

5 A Heuristic Algorithm

First, we will give a theorem to show that the minimum penalty of a polygon with n edges can be found in $O(n)$ time. Before giving the theorem, we have the following lemma.

Lemma 4 *Let the minimum penalty of a traffic network $G = (V, E)$ with traffic light cycle T be P . For a particular vertex $v_1 \in V$, if $t(v_1) = m$, for any m and $0 \leq m \leq T - 1$, there exists a timing setting method whose total penalty is P .*

Proof: Suppose $|V| = n$ and the minimum penalty P can be obtained under the timing setting method $t(v_i) = k_i$, for each $v_i \in V$, $1 \leq i \leq n$. We set that $t(v_i) = (m - k_1 + k_i + T) \bmod T$, $1 \leq i \leq n$, if $t(v_1)$ is changed to m . Suppose v_j is adjacent to v_i . Before the timing setting is changed, the penalty of (v_i, v_j) is $f_T(k_i - d(v_i, v_j) - k_j) + f_T(k_j - d(v_i, v_j) - k_i)$, After the timing setting is changed, the penalty of (v_i, v_j) is $f_T(m - k_1 + k_i - d(v_i, v_j) - (m - k_1 + k_j)) + f_T(m - k_1 + k_j - d(v_i, v_j) - (m - k_1 + k_i)) = f_T(k_i - d(v_i, v_j) - k_j) + f_T(k_j - d(v_i, v_j) - k_i)$. Thus, the total penalty of G is not changed. ■

Theorem 6 *Given a graph $G = (V, E)$, where $V = \{v_i | 1 \leq i \leq n\}$, $E = \{(v_1, v_2), (v_2, v_3), \dots, (v_{n-1}, v_n), (v_n, v_1)\}$, $d_i = d(v_i, v_{i+1})$, $1 \leq i \leq n - 1$ and $d_n = d(v_n, v_1)$, the minimum penalty of G can be found in $O(n)$ time.*

Proof: Without loss of generality, assume that $f_T(2d_n) \geq \max\{f_T(2d_1), f_T(2d_2), \dots, f_T(2d_{n-1})\}$. By Lemma 4, to find the minimum penalty of G , we can first set $t(v_1) = 0$. Let w_{i+1} , $1 \leq i \leq n-1$, denote the number of solutions that (v_i, v_{i+1}) has zero extra penalty when $t(v_i)$ has been set. Let $a_i = (t(v_{i-1}) - d(v_{i-1}, v_i)) \bmod T$, $2 \leq i \leq n$. Note that $t(v_i) = a_i$, $2 \leq i \leq n$, is the first solution that $(v_1, v_2), (v_2, v_3), \dots, (v_{n-1}, v_n)$ have zero extra penalty. By Corollary 1, to find the minimum penalty of G , under the condition that each (v_i, v_{i+1}) , $1 \leq i \leq n-1$, has zero extra penalty, we need only to find the minimum extra penalty of (v_n, v_1) . Let $y_i = w_i - 1$, $2 \leq i \leq n$. To do this, we need only to try to set $t(v_n)$ to each value in the range from a_n to $a_n + \sum_{i=2}^n y_i$. Suppose when $t(v_n) = a_n + (\sum_{i=2}^k y_i) + c$ where $k \leq n$ and $0 \leq c < y_{k+1}$, (v_n, v_1) has the minimum extra penalty. Let $b = (\sum_{i=2}^k y_i) + c$. We set $t(v_2) = (a_2 + y_2) \bmod T$, $t(v_3) = (a_3 + y_2 + y_3) \bmod T$, $\dots$, $t(v_k) = (a_k + \sum_{i=2}^k y_i) \bmod T$, $t(v_{k+1}) = (a_{k+1} + b) \bmod T$, $t(v_{k+2}) = (a_{k+2} + b) \bmod T$, $\dots$, $t(v_n) = (a_n + b) \bmod T$. In this setting, the extra penalty of each edge (v_i, v_{i+1}) , $1 \leq i \leq n-1$, is zero. And, the extra penalty of edge (v_n, v_1) is the minimum. Thus, the penalty of G has been minimized. The above computation can done in $O(n)$ time. In other words, the minimum penalty of G can be found in $O(n)$ time. ■

Now, we will propose a heuristic algorithm with polynomial time to solve the simplified traffic light problem.

Let $G = (V, E)$ be a graph and the distance of each edge $(u, v) \in E$ be $d(u, v)$. Note that the minimum penalty of each edge (u, v) is $f_T(2d(u, v))$. Our heuristic algorithm will find a near minimum penalty of G as follows.

Algorithm 1 *Algorithm Traffic Lights Setting(TLS)*

Input: A traffic light cycle T , where T is an integer and $T \geq 3$, and $G = (V, E)$ where

$V = \{v_i | 1 \leq i \leq n\}$ and every edge e is associated with an integer distance $d(e)$, $0 \leq d(e) \leq T-1$.

Output: $t(v)$ for each $v \in V$ such that the penalty P of G for the simplified traffic light problem is minimized as possible as.

Step 1: Set $t(v_1) = 0$.

Step 2: Set $w(v_i, v_j) = f_T(2d(v_i, v_j))$ as the weight of each edge $(v_i, v_j) \in E$. In G , find a minimum spanning tree rooted at v_1 .

Step 3: $a_i = (t(v_j) - d(v_i, v_j)) \bmod T$ and $b_i = (t(v_j) + d(v_i, v_j)) \bmod T$, where v_j is the father of v_i in the rooted minimum spanning tree.

Step 4: Set $t(v_i) = a_i$, $2 \leq i \leq n$. Let P denote the penalty of G under this setting.

Step 5: For each leaf node v_i do

For each integer x in the range between a_i and b_i .

Let P' be the penalty of G when $t(v_i) = x$.

If $P > P'$ then let $P = P'$ and set $t(v_i) = x$.

end

Algorithm TLS guarantees that each edge on the minimum spanning tree has zero extra penalty. Why does this setting method get a good solution? There are two reasons. First, by Theorem 4, in a polygon(cycle), we can find the minimum penalty in the settings that make all edges, except the one with the largest weight $f_T(2d(e))$, have zero extra penalty. Second, by Lemma 1, for an edge, the sum of the minimum penalty and the maximum penalty is equal to T . In other words, an edge with smaller minimum penalty may have larger extra penalty. For a leaf node v_i on the rooted minimum spanning tree, each integer x in the range between a_i and b_i achieves the goal that each edge on the minimum spanning tree has zero extra penalty. Thus, the job of Step 5 is to do fine tuning to get a better solution. Clearly, the time required for the algorithm is $O(n^2)$, which is spent to find the minimum spanning tree, plus the time needed for Step 5.

The realistic traffic light problem can be easily transformed into the simplified traffic light problem, as done in Section 3.

6 Experiment Results

As noted earlier, it is hard to find the minimum penalty for a traffic network. Thus, we will compare our experiment results of our heuristic algorithm for the simplified traffic light problem with the lower bound of penalty and some other simpler heuristic algorithms.

Consider the traffic network which is a general graph $G = (V, E)$. For each edge (u, v) in G , $f_T(2d(u, v))$ is the minimum penalty of (u, v) . We define the lower bound of the penalty of G as the sum of the minimum penalties of all edges in G .

Table 3 shows the experiment results of seven general graphs of different sizes. In each experiment, shown in one row, 50 cases are tested and the average penalty of them is calculated. The results shown in Table 3 are obtained by four different methods. The second column reports the lower bound by our definition, the third column shows the penalty obtained by algorithm TLS, the fourth column shows the result obtained by tree optimal method and the fifth column shows the penalty when all traffic lights are set randomly. In the tree optimal method, we first find a depth-first spanning tree and then set each edge in the spanning tree to have zero extra penalty. We have to point out that the real minimum penalty should be larger than the lower bound and smaller than our heuristic result.

Table 3 *The experiment results on a general graph consisting of n vertices.*

n	lower bound	TLS	tree optimal	random
10	135.80	195.44 (143.92%)	226.28 (166.63%)	268.52 (197.73%)
25	361.96	527.12 (145.63%)	609.40 (168.36%)	734.84 (203.02%)
50	746.84	1079.20 (144.50%)	1261.44 (168.90%)	1496.36 (200.36%)
80	1184.30	1661.00 (140.25%)	1972.05 (166.52%)	2356.30 (198.96%)
100	1482.50	2083.95 (140.57%)	2461.55 (166.04%)	2976.55 (200.78%)
200	2977.75	4211.90 (141.45%)	4973.80 (167.03%)	5942.60 (199.57%)
500	7505.60	10669.20 (142.15%)	12470.60 (166.15%)	15083.60 (200.96%)

Table 4 shows the experiment results of seven planar graphs of different sizes. In each experiment, shown in one row, 50 cases are tested and the average penalty of them is calculated.

Table 4 *The experiment results on a planar graph consisting of n vertices.*

n	<i>lower bound</i>	<i>TLS</i>	<i>tree optimal</i>	<i>random</i>
10	77.44	91.72 (118.44%)	110.80 (143.08%)	151.32 (195.40%)
25	212.76	250.44 (117.71%)	306.08 (143.86%)	419.88 (197.35%)
50	420.00	504.36 (120.09%)	605.68 (144.21%)	857.68 (204.21%)
80	688.24	820.92 (119.28%)	984.56 (143.05%)	1379.56 (200.45%)
100	835.30	1003.70 (120.16%)	1211.40 (145.03%)	1727.80 (206.85%)
200	1751.00	2082.40 (118.93%)	2434.60 (139.04%)	3436.60 (196.26%)
500	4440.80	5243.60 (118.08%)	6258.40 (140.93%)	8716.00 (196.27%)

7 Conclusions

In a traffic network, the importance for different roads is different. Thus, another factor for setting traffic lights is the traffic flow on the roads. When the traffic flow is involved, the problem becomes the weighted traffic light problem. Now, we give a formal definition for the weighted traffic light problem. Given a traffic light cycle T , where T is even, and a graph $G = (V, E)$ in which each edge (u, v) is associated with a distance $d(u, v)$ and the traffic flow $w(u, v)$, we want to assign $t(v)$ for each vertex $v \in V$. Let $P_{uv} = \min\{c_{uv}, T - c_{uv}\}$, where $c_{uv} = |t(v) - d(u, v) - t(u)| \bmod T$ if $(u, v) \in E$ and $c_{uv} = 0$ if $(u, v) \notin E$. Our goal is to assign $t(v)$ for each $v \in V$ such that the total penalty $(\sum_{u \in V} \sum_{v \in V} w(u, v) \cdot P_{uv})$ is minimized. This weighted traffic light problem can also be solved by our heuristic algorithm. For the weighted traffic light problem, if every traffic light can get only the traffic flow of neighboring vertices, an efficient distributed algorithm to find the minimum penalty of a traffic network may be preferred.

In our traffic light problem, we find the total minimum penalty of a traffic network. In fact, a driver cares how to arrive at the destination as soon as possible. Consider a graph $G = (V, E)$, where $V = \{v_1, v_2, v_3, v_4, v_5\}$ and $E = \{(v_i, v_{i+1}) \mid 1 \leq i \leq 4\}$ (G is a path of 5 vertices) and $d(v_1, v_2) = 3$, $d(v_2, v_3) = 1$, $d(v_3, v_4) = 4$, $d(v_4, v_5) = 5$. If the traffic light cycle is 20, then we can get the minimum penalty of G , which is 26, when $t(v_1) = 0$, $t(v_2) = 3$, $t(v_3) = 4$, $t(v_4) = 8$ and $t(v_5) = 13$. In fact, if a driver wants to drive from v_1 to v_5 , then the driver needs to wait only 10 unit of times at most, no matter what time the driver starts from v_1 . And if a driver drives from v_5 to v_1 , then the driver needs to wait 10 unit of times

at most, too. Thus, there may be another way to model the traffic light problem. And an algorithm is needed to be designed to get the shortest time from some special source points to some special destination points in a traffic network.

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